Department of Economics
University of Arkansas
Fall 2021

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ECON 5783 - Applied Microeconometrics
1.00 - 2.15PM Mon/Wed, WJWH103

This course covers the principles of causal inference. Methods include panel data models, instrumental variables, regression discontinuity designs, difference-in-differences, and matching. Emphasis on developing a solid understanding of the underlying econometric principles of the methods taught as well as on their empirical application.

Textbooks

The main (and only required) textbook for the course is Angrist and Pischke's *Mastering Metrics* (2015). At the same time, I highly recommend their more advanced version of the book *Mostly Harmless Econometrics* (2009) that covers similar topics but with greater detail and more mathematical rigor. This more advanced book is a worthwhile reference for those who want to do further empirical research — be that in academia, private sector, or government — using causal inference methods.

Grading Policy

Grading for the course will be based on problem sets (25%), exams (40%), and the final project (35%). Letter grades will be assigned based on the total score. All students are expected to complete all problem sets and exams. Don't expect extra-credit assignments.

Problem sets will usually due two weeks after assigned. The problem sets will require you to analyze datasets using techniques discussed in class. Students are encouraged to work in groups to discuss how the approach the problem sets, but each student must hand in his or her own set of answers. Missing or late problem sets will receive no credit. Problem sets would require Stata. A session will be held to familiarize students with Stata.

Exams will be held during class time on October 11th and November 18th.

For the **final project**, you will write an empirical paper using microeconometric data. A complete project must include a research question in the form of "the impact of X on Y", descriptions of context, empirical strategy (including why the chosen strategy is valid for the question/data), results, and a brief discussion on limitations. You also will need to present a preliminary draft in the last 3 weeks of class.

Non-PhD economics students: (i) may work in a group of two; and (ii) are recommended but not required to include a brief literature review. **PhD economics students** should submit a single-authored paper and include a brief literature review.

Tentative Schedule

Week	Topic	Readings
1	Introduction & Probability Basics	
2	Randomized Controlled Trial	Angrist & Pischke (2015), Ch. 1
3-5	Linear Models & Matching	Angrist & Pischke (2015), Ch. 2, Caliendo and Kopeinig (2008), "Some Practical Guidance for the Implementation of Propensity Score Matching"
6-7	Instrumental Variable	Angrist & Pischke (2015), Ch. 3
8	— **Exam 1 (10/11)** —	
8-9	Regression Discontinuity Design	Angrist & Pischke (2015), Ch. 4
10-11	Panel Data and Difference in Differences	Angrist & Pischke (2015), Ch. 5
12	Discuss project ideas	
13	— **Exam 2 (11/17)** —	
14-16	Project presentations	

Note: All required readings from Angrist & Pischke (2015) include the appendix

Covid Protocol in Classrooms

Everyone is expected to follow the current university health protocols related to Covid. As of the writing of this syllabus, masks must be worn indoors. See here for more information and updates.

Academic Dishonesty

Students are expected to adhere to the University's policy regarding academic honesty. It is explained in the Catalog of Studies, and can be found online at: http://catalogofstudies.uark.edu/1213-Undergraduate_Catalog-1.pdf

Accommodations for Students with Disabilities

University of Arkansas Academic Policy requires that students with disabilities are provided reasonable accommodations to ensure their equal access to course content. If you have a documented disability and require accommodations, please contact me privately at the beginning of the semester to make arrangements for necessary classroom adjustments. Please note, you must first verify your eligibility for these through the Center for Educational Access (contact 479-575-3104 or visit http://cea.uark.edu for more information on registration procedures).

Inclement Weather Policy

In case of severe weather, a notification will be sent out via Blackboard regarding cancellation of any class. Further instructions regarding the make-up of missed class will also be posted on the Blackboard site.